

day 38 prep

first we need a way to produce something like a square wave or a saw-tooth wave.

we also need to go through the math of fourier transforms. then go into fast fourier transforms, then on to doing some fourier transforms of simple things and then more complex things.

the mod function works great here and it is a real pro tip.

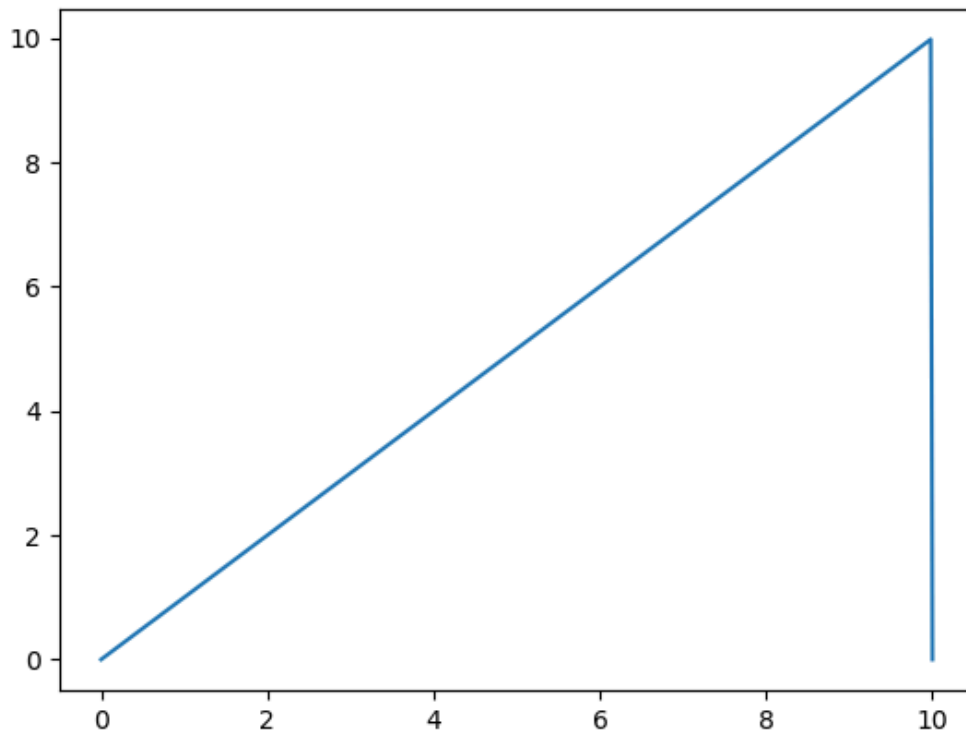
```
In [43]: L = 10
x = np.linspace(0, L, 500)  # show three periods bc it looks cooler
y = (x % L)                 # periodic sawtooth

fig0,ax0=plt.subplots()
ax0.plot(x, y)
```

```
/tmp/ipykernel_15176/3997010046.py:5: RuntimeWarning: More than 20 figures h
ave been opened. Figures created through the pyplot interface (`matplotlib.p
yplot.figure`) are retained until explicitly closed and may consume too much
memory. (To control this warning, see the rcParam `figure.max_open_warning
`). Consider using `matplotlib.pyplot.close()`.
  fig0,ax0=plt.subplots()
```

```
Out[43]: [<matplotlib.lines.Line2D at 0x7252455f7e30>]
```

Figure

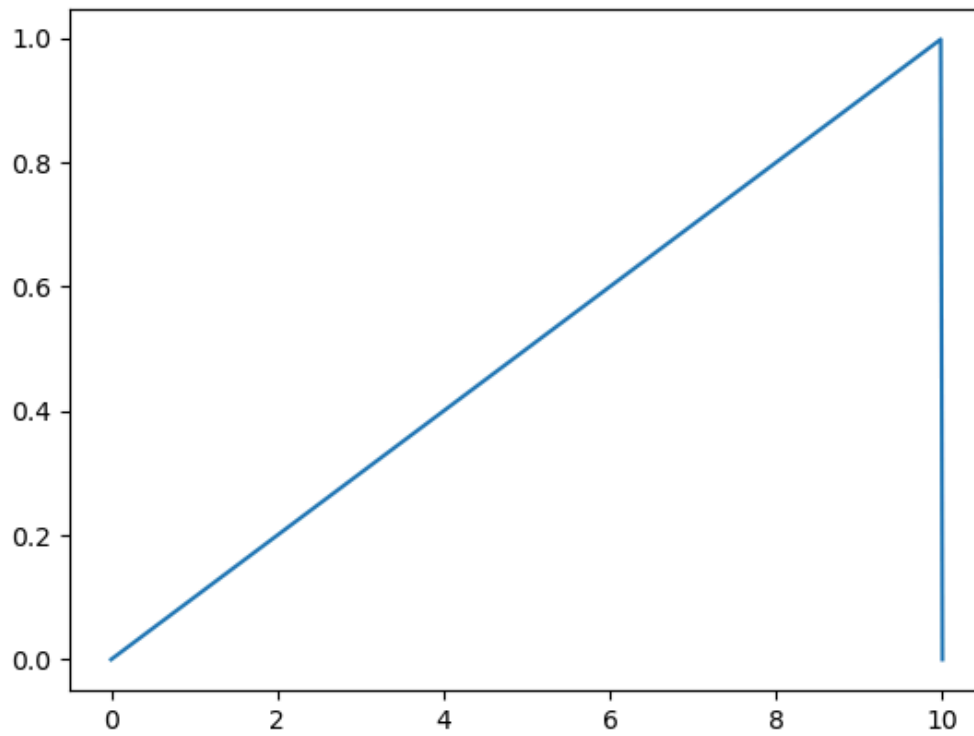


so how can i make a sawtooth of that quickly.

```
In [44]: def saw(x, L=2*np.pi):  
         return (x % L)/L    # normalized to [0,1]  
  
         fig1, ax1 = plt.subplots()  
         ax1.plot(x,saw(x, 10))
```

```
Out[44]: [<matplotlib.lines.Line2D at 0x725245338ef0>]
```

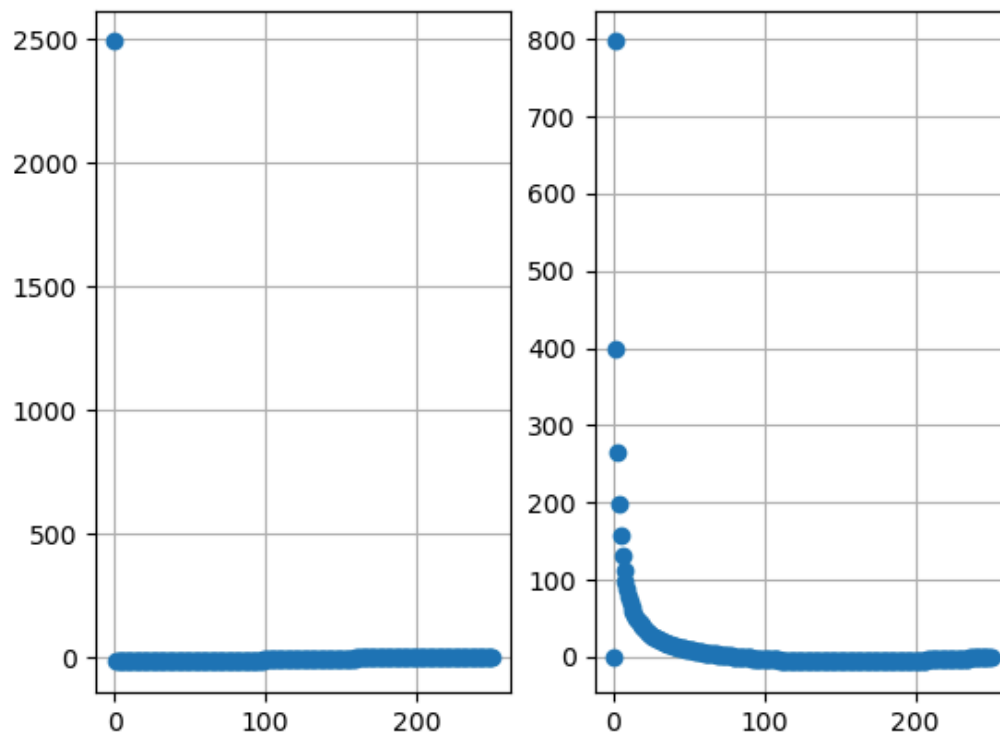
Figure



```
In [59]: c = np.fft.rfft(y)
y1 = np.fft.irfft(c)
c2 = np.fft.fft(y)
y2 = np.fft.ifft(c2)
```

```
In [60]: fig2, (ax2, ax3) = plt.subplots(1,2)
ax2.plot(c.real, 'o')
ax3.plot(c.imag, 'o')
ax2.grid()
ax3.grid()
```

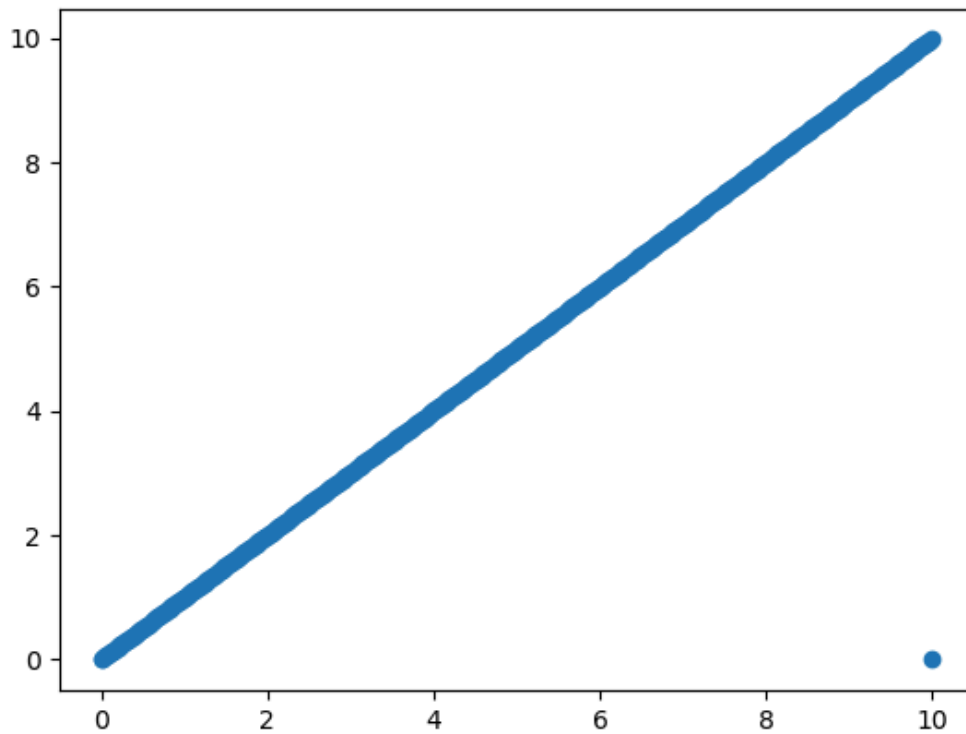
Figure



```
In [61]: fig3, ax3 = plt.subplots()
         ax3.plot(x,y1, 'o')
```

```
Out[61]: [<matplotlib.lines.Line2D at 0x72523f882600>]
```

Figure



```
In [42]: c[3]
```

```
Out[42]: (-5.500000000000002+4.765777128986844j)
```

```
In [58]: print(c[:10])
          print(c2[:10])
```

```
[2490.      +0.j      -15.00923048+797.23330098j
-15.00686193+398.41244009j -15.00291477+265.38143667j
-14.99738961+198.7979468j -14.99028732+158.79351742j
-14.98160904+132.07867727j -14.97135613+112.95794988j
-14.95953021 +98.58360681j -14.94613315 +87.3735854j ]
[2490.      +0.j      -15.00923048+797.23330098j
-15.00686193+398.41244009j -15.00291477+265.38143667j
-14.99738961+198.7979468j -14.99028732+158.79351742j
-14.98160904+132.07867727j -14.97135613+112.95794988j
-14.95953021 +98.58360681j -14.94613315 +87.3735854j ]
```

```
In [ ]:
```